

/CTDI/ Repair Print Framework

End-User Manual & Comprehensive Operational Guide

DOCUMENT CONTROL & SYSTEM SCOPE

- **Document Version:** 2.1 (Production Release)
- **System Namespace:** /CTDI/
- **Target Roles:** Workshop Technicians, Dispatchers, Warehouse Clerks, Repair Administrators, Key Users & Service Managers
- **Target SAP Releases:** SAP ERP 6.0 (EHP7/EHP8), SAP S/4HANA (On-Premise 1809+)
- **Included Reports:** /CTDI/PRINT_REPAIR (Single Print), /CTDI/PRINT_REPAIR_MASS (ALV Mass Workplace)
- **Publication Date:** September 2026

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1. Framework Overview & Value Proposition

The **/CTDI/ Dynamic Repair Print Framework** is SAP-native software designed to handle all printing and PDF export requirements for customer repair orders, service notifications, and warranties.

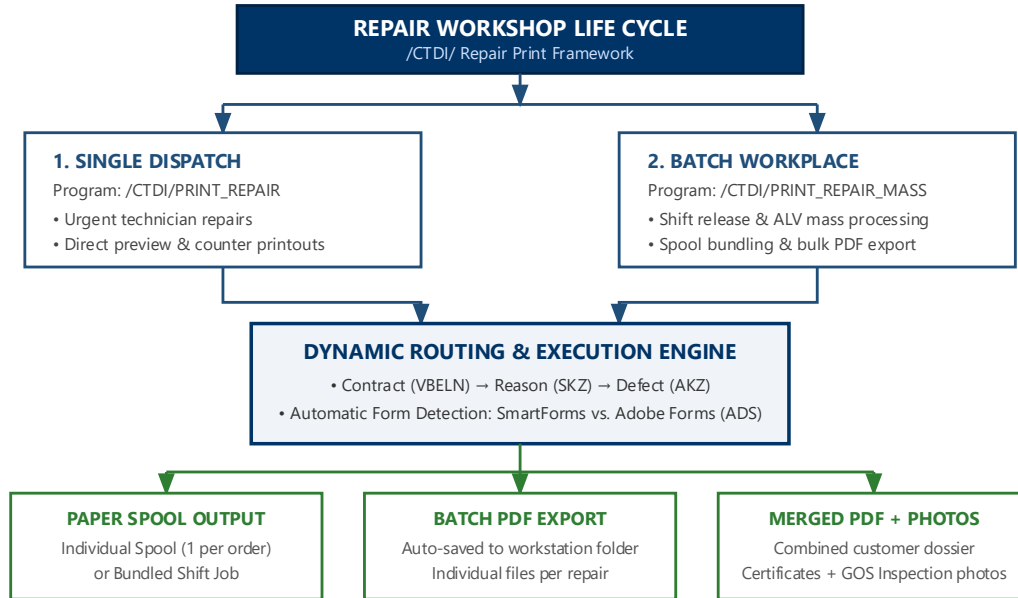


Figure: Repair Workshop Life Cycle

Key Advantages for the Daily Operation

- 1. Automatic Form Layout Detection:** You do not need to choose whether a customer uses **SmartForms** or modern **Adobe Interactive Forms (ADS)**. The system identifies the layout and renders it automatically.
- 2. Integrated Inspection & Defect Photos:** Any photos attached via SAP GOS (Generic Object Services) or Content Server / ArchiveLink are automatically scaled to DIN A4 and appended to the PDF output.
- 3. Flexible Spool Bundling:** Instead of flooding the printer with hundreds of 1-page spool requests, you can bundle an entire shift's batch into a single spool job.
- 4. Smart Folder PDF Downloads:** When downloading PDFs for multiple orders, the system prompts you for the destination directory once and saves all files quietly in the background without repeated dialogs.
- 5. Direct Master Data Drilldown:** Double-click on any order, notification, or contract in the mass screen to jump straight to standard transactions (**IW33**, **IW53**, **VA43**, **VA03**).

2. User Roles & Transaction Overview

Role	Main Transactions	Primary Tasks
Workshop Technician / Operator	/CTDI/PRINT_REPAIR	Print single repair slip, preview output, check inspection photos.
Dispatcher / Shipping Clerk	/CTDI/PRINT_REPAIR_MASS	Review ready orders, print batch spools, export merged customer dossiers.
Key User / Repair Supervisor	SM30 (/CTDI/REP_FORMS, /CTDI/REP_RESULT)	Configure customer layouts, maintain repair outcome texts, adjust image rules.
IT & Spool Administrator	SP01, SLG1, NACE	Spool queue management, application log analysis, output condition records.

3. Single Repair Order Printing (`/CTDI/PRINT\_REPAIR`)

Use `/CTDI/PRINT\_REPAIR` when working on an individual repair order at the service desk, packing station, or technician workbench.

3.1 Selection Screen & Parameters

Figure: Single Repair Order Printout Selection Screen

Field Explanations

Field	Description	Mandatory?	Notes / Recommendation
Repair / Order ID (p_aufnr)	The SAP PM/CS Service Order number (e.g., 4001234).	Yes	Leading zeros are added automatically by SAP.
Serial Number (p_sernr)	Specific equipment serial number.	No	Leave blank if printing for the primary order equipment.
Append attached images (p_images)	Appends defect/inspection photos to the PDF.	No	Check this if the customer requires photographic evidence.

i NOTE

When executing from standard transaction `IW42` (Overall Completion Confirmation), the system defaults directly to printer spool mode. When executing manually from the selection screen, the system defaults to generating a **local PDF file**.

3.2 Step-by-Step: Printing a Single Order to Paper

1. Enter Transaction code `/CTDI/PRINT_REPAIR` in the SAP command field.
 2. Enter the **Repair Order ID** in `p_aufnr`.
 3. If paper printout is required, ensure the output device configured in your SAP User Profile (**SU3** & rarr; *Defaults* & rarr; *Output Device*) is set to your physical workstation printer.
 4. Press `F8` (Execute).
 5. The system resolves the layout and dispatches the document directly to the printer spool. A status message appears in the bottom status bar:
- > ✓ **Print driver completed successfully for Repair 4001234**

3.3 Step-by-Step: Generating & Saving a PDF with Inspection Photos

1. Launch `/CTDI/PRINT_REPAIR`.
2. Enter the **Repair Order ID** (e.g., **4001234**).
3. Check the box ☒ **Append attached images to PDF**.
4. Press `F8` (Execute).
5. A Windows **File Save Dialog** opens automatically:
 - File Name suggestion: **Repair_4001234_<Timestamp>.pdf**
 - Choose your target local folder (e.g., **C:\Repairs\Reports**) and click **Save**.
6. The resulting PDF document contains:
 - **Page 1..N**: The official Repair Delivery Note / Certificate (Adobe Form or SmartForm).
 - **Page N+1..**: All attached inspection and damage photos, scaled to A4 with aspect ratio preservation and timestamp captions.

3.4 Quick-Access Customizing Toolbar

At the top of the selection screen, authorized Key Users and Supervisors have direct access to system customizing tables without leaving the transaction:

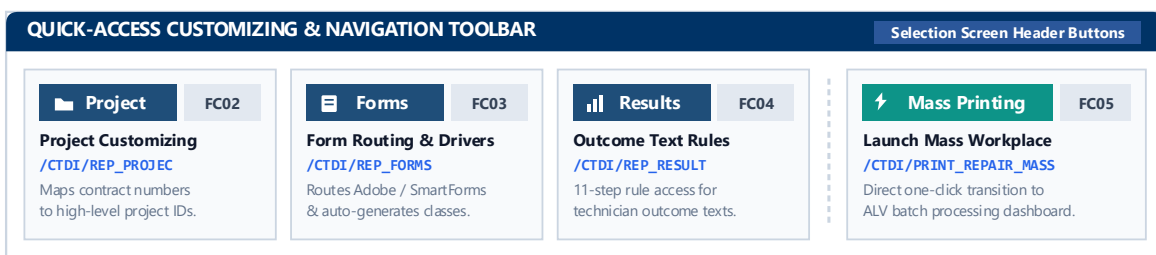


Figure: Quick-Access Customizing Toolbar

- `[Project]` (FC02): Opens Table Maintenance for `/CTDI/REP_PROJEC` (Contract-to-project mappings).

- **`[Forms]` (FC03):** Opens Table Maintenance for **/CTDI/REP_FORMS** (Contract-to-form & driver assignments).
- **`[Results]` (FC04):** Opens Table Maintenance for **/CTDI/REP_RESULT** (11-step repair outcome text mappings).
- **`[Mass Printing]` (FC05):** Instantly launches the Mass Processing Workplace (**/CTDI/PRINT_REPAIR_MASS**).

 IMPORTANT

The customizing buttons **[Project]**, **[Forms]**, and **[Results]** are only visible if your SAP user ID is granted maintenance authorization in table **ZSM30_USER**. Regular operators will only see the **[Mass Printing]** shortcut.

4. Mass Printing & Processing Workplace (`/CTDI/PRINT\_REPAIR\_MASS`)

`/CTDI/PRINT\_REPAIR\_MASS` is the primary operational dashboard for dispatchers, shipping clerks, and repair supervisors. It displays an interactive ALV grid containing all ready repair orders and allows bulk execution.

4.1 Selection Screen & Filtering Orders

Figure: Mass Print Selection Screen

1. Enter your search criteria (Order range, Plant, or Date range).
2. If you want inspection photos included in PDF exports, select **[X] Append attached images to PDF**.
3. Press **`F8`** (Execute).

4.2 The Interactive ALV Workplace Grid

The program retrieves all matching orders, deduplicates records to guarantee **strictly one row per repair order**, pre-evaluates form layouts, and renders the interactive ALV Workplace:

Mass Print Repair Orders — Interactive ALV Workplace (42 Orders Found • Deduplicated 1 Row / Order)										
Print Selected		Save as PDF		PDF Merge		Preview		Pool Mode: BUNDLED		Attach: ON
										Render: RAW PDF
Stat	Att.	Order ID	Customer PO	Contract	ZZ Notif.	SKZ	AKZ	Form Tech	Form Name	Execution Message
●		4001001	50008812	40000010	10005541	10	SCRP	Adobe (ADS)	/CTDI/REP_D	✓ Printed to Spool Job #140221 2 inspection photos appended
●		4001002	50008813	40000010	10005542	20	REPR	Adobe (ADS)	/CTDI/REP_D	✓ Exported to PDF successfully Saved in C:\Repairs\2026-09-21\
○	—	4001003	50008814	40000025	10005543	10	SWAP	SmartForms	/CTDI/REP_SF	Ready for processing
●		4001004	50008815	40000025	10005544	30	NFF	SmartForms	/CTDI/REP_SF	⚠ Warning: Global fallback used No contract specific rule in REP_FORMS
ALV Workplace Quick Guide: • Double-Click Navigation: Click Order ID → IW33 Notification → IW53 Contract → VA43 Sales Order → VA03 • Attachment Icons: Confirmed Content Server defect photos (ZRS.JPG) General GOS attachments • Batch Execution: Select target rows with checkboxes, then click [Print Selected], [Save as PDF], or [PDF Merge].										
4 Orders displayed (of 42 loaded) Filter: Plant 1000, Date 01.09.2026 - 21.09.2026 Layout: /DEFAULT Interactive SAP ALV Grid Workplace										

Figure: Interactive ALV Workplace Grid

4.3 Understanding Attachment & Status Indicators

Column: `Att.` (`ICON\_IMG`) — Attachment Status

Icon	SAP Symbol	Meaning	What it means for you
	@9T@ (icon_bmp)	Confirmed Image Attachments	High-resolution photos exist in the Content Server / ArchiveLink (ZRS.JPG). These will be appended.
	@1S@ (icon_attachment)	Standard GOS Attachments	Office/GOS attachments exist on the order or linked notification. Image formats (JPG, PNG, etc.) will be appended.
Blank	Empty	No Attachments Found	Order has no linked photos or files. Standard form output only.

Column: `Stat` (`ICON`) — Execution Result

Icon	SAP Symbol	State	Explanation
○	@BZ@ (icon_led_inactive)	Ready / Untouched	Order has been loaded but not yet printed in this session.
●	@5B@ (icon_led_green)	Success	Document printed to spool or successfully downloaded as PDF.

Icon	SAP Symbol	State	Explanation
	@5D@ (icon_led_yellow)	Warning / Fallback	Processed with warnings, or non-critical fallback configuration used.
	@5C@ (icon_led_red)	Error	Execution failed. See the Message column for root-cause details.

4.4 1-Click Drilldown Navigation

You can double-click directly on specific cell values in the ALV to open the corresponding SAP master transaction in a new session:

Column Clicked	Target Transaction	Description
Order ID (AUFNR)	`IW33`	Display PM/CS Repair Order (operations, components, costs).
Z2 Notif. (QMNUM)	`IW53`	Display Service Notification (reported defect, customer remarks).
Contract (CONTRACT_ID)	`VA43`	Display Service / Sales Contract (SLA conditions, validity).
Sales Order (KDAUF)	`VA03`	Display Sales Order details.

4.5 Spool Printing Modes (Individual, Bundled, Merged)

When clicking **[Print Selected]** (**PRINT_SEL**), the framework routes documents according to the currently active **Spool Mode**. You can switch the mode at any time by clicking the **[Spool Mode]** button in the ALV toolbar.

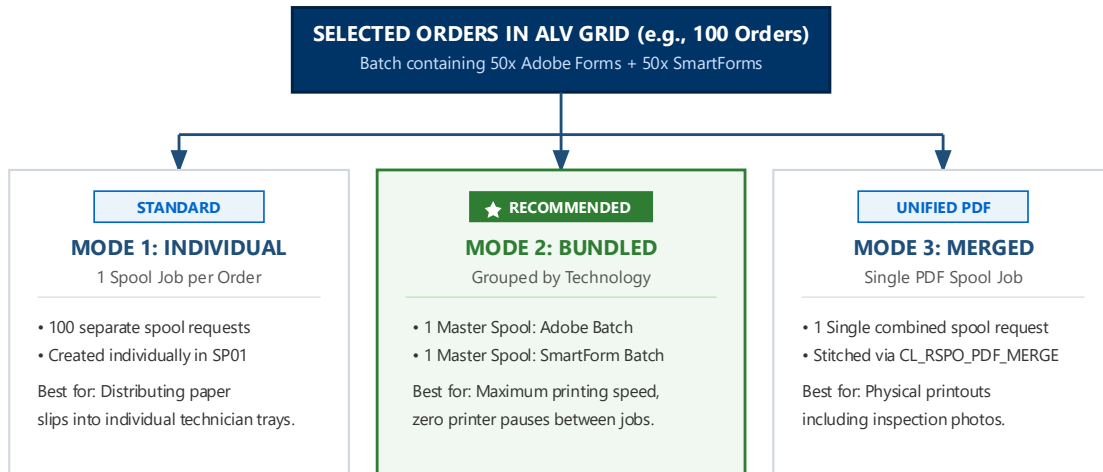


Figure: Comparison of Spool Modes

1. Individual Mode (`INDIVIDUAL`):

- Creates a distinct spool job for every order.
- **Best for:** When printed paper must be immediately separated into 100 physical technician trays by different workers.

2. Bundled Mode (`BUNDLED`) — RECOMMENDED:

- Uses high-speed SAP spool bundling (**FP_JOB_OPEN** / **SSF_OPEN** simple bundling).
- Generates only 1 or 2 large spool jobs (e.g., **Mass Print Adobe 21.09.2026_14:30**).
- **Best for:** Maximum printing speed. Eliminates printer pauses between single jobs.

3. Merged Mode (`MERGED`):

- Converts all documents into PDF and merges them into one single spool document via **CL_RSPO_PDF_MERGE**.
- **Best for:** When inspection photos must be included directly in the paper spool output.

4.6 PDF Export Modes (Batch Download vs. Merged PDF)

The ALV toolbar provides two distinct options for creating PDF files on your computer:

Option A: `[Save as PDF]` (`PDF\_SEL`) — Individual File Batch

- **What it does:** Generates a separate **.pdf** file for each selected order.
- **Smart Directory Memory:**
 1. On the first order, SAP asks you where to save the files (e.g., **D:\Shift_Export**).
 2. For all remaining selected orders, the system automatically saves into that same directory without prompting again!
 3. File naming format: **Repair_<OrderNumber>_<Timestamp>.pdf**.

Option B: `[PDF Merge]` (`PDF\_MERGE`) — Single Combined Document

- **What it does:** Renders all selected repair certificates, appends all corresponding inspection photos, and stitches everything into **one single master PDF dossier**.

- Prompts you once for the output file name (e.g., **Repair-Merged-25-Orders-20260921.pdf**).
 - **Best for:** Sending a complete bulk repair dossier to an external customer or archiving team via email.
-

4.7 On-Screen Print Preview

To visually inspect a document layout before sending it to the printer:

1. Select a single row in the ALV table.
 2. Click **[Preview]** (**PREVIEW**).
 3. The SAP Print Preview window opens immediately, showing the exact rendered form with live database values.
 4. Close the preview window to return to the ALV grid without generating a spool job.
-

5. Inspection Photos & GOS Image Attachment Guide

The **/CTDI/** Print Framework features an automated photo append pipeline managed by **/CTDI/CL_PRINT_GOS_IMAGES**.

5.1 Where Images Come From

The system queries two standard SAP attachment locations:

1. Generic Object Services (GOS / SOFFICE):

- Attachments stored directly on the Service Order (**BUS2007**) via the standard GOS paperclip icon.
- Attachments stored on the linked Service Notification (**QMEL / BUS2078**).

2. SAP Content Server / ArchiveLink (`TOA01`):

- Professional defect inspection images archived under document type **ZRS_JPG**.

5.2 Supported Image Formats

File Extension	Format	Support Status	Notes
.JPG / .JPEG	JPEG	☒ Full Native	Full EXIF orientation support, auto-scaled.
.PNG	Portable Network Graphics	☒ Full Native	Transparency preserved against white background.
.BMP	Windows Bitmap	☒ Full Native	Converted and scaled.
.TIF / .TIFF	Tagged Image File	☒ Full Native	Standard single-page inspection TIFFs supported.
.PDF / .DOCX	Non-Image Documents	☐ Preserved	Filtered out from image pipeline (not converted to photos).

5.3 Automatic Page Layout & Aspect Ratio Preservation

Attached photos are never stretched or distorted. The engine calculates the optimal bounding box and packs **up to 2 photos per DIN A4 page** vertically:

DIN A4 PORTRAIT PAGE

Photo 1 (Inspection Front)

Centered Defect Photo

(Aspect Ratio Preserved)

Photo 2 (Serial Plate Macro)

Damage Detail

(Aspect Ratio Preserved)

AUTOMATIC DIN A4 PHOTO PACKING ALGORITHM

- Exact DIN A4 Portrait dimensions (595 × 842 pt, 36 pt margins)
- Packs up to 2 inspection photos per A4 page vertically
- Strict aspect-ratio preservation — zero distortion or stretching
- Automatic horizontal centering inside each image slot
- Metadata headers included on each photo (Order ID + Caption)
- Multi-page handling: Images 3, 4, etc. automatically flow to Page 2+

Supported Graphic Formats:

✓ JPEG / JPG

✓ PNG (Transparency)

✓ BMP

✓ TIFF / TIF

Figure: Inspection Photo Layout

- If an order has **1 image**: It is centered gracefully on the page.
- If an order has **2 images**: Both are stacked vertically on Page 1.
- If an order has **3+ images**: A new A4 page is opened automatically for images 3 & 4.

5.4 Rendering Engines: Adobe ADS vs. Built-in Raw PDF

In the ALV Toolbar, you can toggle the image rendering method using `[Img Render]` (**IMG_RENDER**):

Mode	Technology	When to use
Raw PDF (R)	Built-in Lightweight Vector Engine	Default & Recommended. Fastest execution, zero server overhead, works even if Adobe Document Services (ADS) is temporarily offline.
ADS Form (A)	Adobe Document Services (<code>/CTDI/REPAIR_IMG</code>)	Uses SAP standard ADS template rendering. Ideal for high-end color profile matching.

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6. Key User & Supervisor Customizing Guide

Key users can maintain customer form routing and repair texts via Transaction `SM30` (or by clicking the quick-access buttons on the **/CTDI/PRINT_REPAIR** selection screen).

6.1 Form Routing Hierarchy (`/CTDI/REP\_FORMS`)

When a repair order is processed, the framework evaluates table `CTDI/REP\_FORMS` using an **8-step priority fallback sequence**:

8-STEP FORM ROUTING PRIORITY HIERARCHY (/CTDI/REP_FORMS)				
Priority	Contract (VBELN)	Reason (SKZ)	Defect (AKZ)	Match Granularity
Step 1	Specific Contract	Specific Reason	Specific Defect	Highest Specificity
Step 2	Specific Contract	Specific Reason	(Blank / Any)	Contract + Reason
Step 3	Specific Contract	(Blank / Any)	Specific Defect	Contract + Defect
Step 4	Specific Contract	(Blank / Any)	(Blank / Any)	Contract Default (Fallback)
Step 5	(Blank / Global)	Specific Reason	Specific Defect	Global + Reason + Defect
Step 6	(Blank / Global)	Specific Reason	(Blank / Any)	Global + Reason
Step 7	(Blank / Global)	(Blank / Any)	Specific Defect	Global + Defect
Step 8	(Blank / Global)	(Blank / Any)	(Blank / Any)	Global System Fallback

Evaluation stops immediately at the first matching record. Buffered in-memory for fast batch execution.

Figure: Form Routing Priority Hierarchy

Table Columns in `CTDI/REP\_FORMS`

Column	Key?	Description	Example
VBELN	Key	Sales Contract / Customer Project Number	40000010 (or blank for global)
SKZ	Key	Confirmation Reason (AFRU-BEMOT)	10 (Standard Repair)
AKZ	Key	Defect / Reason Code (QMEL-QMCD)	SCRP (Scrap)
FORM_NAME		SmartForm or Adobe Form layout name	/CTDI/REPAIR_D
CLASS_NAME		Driver class inheriting from /CTDI/CL_PRINT_DRIVER_BASE	Leave blank for auto-generation!
APPEND_IMAGES		Default rule for attaching inspection photos	X = Always attach, = Do not attach



TIP

Automatic Class Generator: When creating a new contract entry in SM30 for **/CTDI/REP_FORMS**, you can leave **CLASS_NAME** empty. The system will automatically detect this and ask:

"Do you want to generate a new driver class /CTDI/CL_PRINT_DRIVER_<Contract>?"

Clicking **Yes** automatically creates and activates a clean ABAP class ready for custom logic!

6.2 Repair Outcome Text Resolution (`/CTDI/REP\_RESULT`)

Table ``/CTDI/REP_RESULT`` controls the customer-facing outcome text printed on the certificate (e.g., *"Replaced main board and recalibrated antenna"*). It uses an **11-step rule access sequence** evaluating Contract, SKZ, AKZ, and whether a device exchange (Tauschfall) took place.

6.3 Project Definitions (`/CTDI/REP\_PROJEC`)

Table ``/CTDI/REP_PROJEC`` maps the high-level business project name and customer identifier to the SAP Sales Contract number.

7. Troubleshooting, Diagnostics & FAQ

7.1 Spool Jobs in `SP01`

To check the status of physical print jobs:

1. Open Transaction **`SP01`**.
2. Filter by your User Name (**sy-uname**) and Date.
3. Look for the standardized Job Titles:
 - **Mass Print Adobe <Date>_<Time>** &arr; Bundled Adobe PDF job.
 - **Mass Print SmartForms <Date>_<Time>** &arr; Bundled SmartForms job.
 - **Mass Print Merged <Date>_<Time>** &arr; Unified merged spool job.
4. Status **COMPLETED** indicates the document reached the physical printer.

7.2 Application Logs in `SLG1`

If an order fails with a red LED in the ALV:

1. Open Transaction **`SLG1`**.
2. Enter Object: **`/CTDI/PRINT\_REPAIR`**.
3. Press **`F8`** (Execute).
4. Full diagnostic call stacks, parameter dumps, and root exceptions are logged for IT review.

7.3 Frequently Encountered Issues & Instant Fixes

Issue 1: ALV row turns Yellow with message **"No configuration found in /CTDI/REP_FORMS"**

- **Cause:** No active entry in **/CTDI/REP_FORMS** matches the Contract, SKZ, or AKZ of this order, and no Global Fallback (blank contract entry) is defined.
- **Solution:** Click the **[Forms]** button on the selection screen, add an entry for the contract (or a global blank entry), and enter the desired form name.

Issue 2: Spool error **"Printer destination invalid or USR01 default missing"**

- **Cause:** Your user profile has no default printer configured.
- **Solution:** Go to Transaction **`SU3`** &arr; Tab **`Defaults`** &arr; Set **Output Device** (e.g. **LOCL** or your network printer) &arr; Check **Output Immediately** &arr; Save (**Ctrl+S**).

Issue 3: Error **"PDF merge initialization failed (ADS connection error)"**

- **Cause:** The SAP Adobe Document Services RFC connection (**ADS**) is temporarily down.
- **Solution:** In the ALV toolbar, click **`[Img Render]`** and switch to **Raw PDF** mode. Raw PDF does not require ADS and runs completely independently.

Issue 4: Photos attached in GOS do not appear in the PDF

- **Cause:** The checkbox *Append attached images* was unchecked, or the attached files are non-image formats (e.g., Word **.docx** or Excel **.xlsx**).
 - **Solution:** Verify the attachment is a valid image (**.jpg**, **.png**, **.bmp**, **.tif**) and ensure the attachment toggle is active (**[Attach: ON]**).
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8. Quick Reference Cheat Sheet

I want to...	Action to take
Print 1 order immediately	Run /CTDI/PRINT_REPAIR , enter Order ID, hit F8 .
Print 50 orders to paper without stopping	Run /CTDI/PRINT_REPAIR_MASS , select rows, click [Print Selected] with Spool Mode set to Bundled .
Send 1 combined PDF of 20 orders to a customer	Run /CTDI/PRINT_REPAIR_MASS , select rows, click [PDF Merge] .
Export 100 individual PDFs into a folder	Run /CTDI/PRINT_REPAIR_MASS , select rows, click [Save as PDF] . Choose the folder once; rest is automatic.
Inspect an order before printing	Double-click the Order ID in the ALV to open IW33 , or click [Preview] .
Check if an order has photos	Look at the Att. column:  = Content Server photos,  = GOS files.
Switch Spool Mode	Click [Spool Mode] on the ALV toolbar (choose Individual, Bundled, or Merged).
Add a new customer form	Open /CTDI/REP_FORMS via SM30 , add the Contract number and Form Name, leave Class blank to auto-generate.

/CTDI/ Repair Print Framework — Built with Precision for SAP Service Operations.